

SPECIAL CHARACTERISTICS CLASSIFICATION GUIDE

AIAG APQP | IATF 16949 Clause 8.3.3.1 | Every SC must be cascaded: Drawing → DFMEA → PFMEA → Control Plan

CC

CHARACTERISTIC CRITICAL

Could affect SAFETY or NON-COMPLIANCE with government regulations

EXAMPLES:

- Brake system hydraulic pressure
- Airbag sensor output voltage
- Steering column torque
- Fuel system burst pressure
- Seatbelt load capacity
- Structural weld strength (safety joints)

CONTROL REQUIREMENTS:

- ✓ Cpk ≥ 1.67 mandatory (preliminary)
- ✓ 100% inspection if Cpk not achieved
- ✓ GRR < 10% for all gauges
- ✓ SPC charts running at all times
- ✓ Reaction plan: STOP production
- ✓ Must appear on: DFMEA, PFMEA, CP
- ✓ PPAP evidence mandatory

SC

CHARACTERISTIC SIGNIFICANT

Could significantly affect FIT, FUNCTION, or CUSTOMER SATISFACTION

EXAMPLES:

- Oil seal surface finish (Ra)
- Gear mesh dimension
- Sealing surface flatness
- Thread engagement length
- Bearing clearance
- Interference fit dimensions

CONTROL REQUIREMENTS:

- ✓ Cpk ≥ 1.67 preferred, min 1.33
- ✓ Statistical process control (SPC)
- ✓ GRR < 10% for SC gauges
- ✓ Reaction plan: Contain + Notify QE
- ✓ Must appear on PFMEA and CP
- ✓ PPAP: Include capability evidence
- ✓ Operator awareness training required

OTHER

SPECIFIC CHARS CUSTOMER-

Defined by each customer — check CSRs. Examples: KCC (GM), CC/SC (Ford), CTQ (Toyota-style)

EXAMPLES:

- GM KCC: Key Control Characteristic
- Ford CC: Critical Characteristic
- Ford SC: Significant Characteristic
- Toyota: Process Characteristics (PC)
- Honda: Safety Characteristic (S/C)
- Maruti: 'A' and 'B' class characteristics

CONTROL REQUIREMENTS:

- ✓ Follow customer-specific requirements
- ✓ Check AIAG CSR portal for latest
- ✓ Same rigor as CC/SC in most cases
- ✓ Customer audit will verify these
- ✓ Cross-reference CSR with Control Plan
- ✓ Training: All QE must know customer symbols